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# Can You Stop Elly De La Cruz From Stealing Second Base?

## 1. Background

In this project, I explored the possibility of preventing Elly De La Cruz, the stolen base leader of the 2024 MLB season, and arguably one of the fastest players in the league, from stealing second base. First, I identified the factors that influence a stolen base attempt. The most important variables were a pitcher's time to the plate, the time for the ball to reach home, the catcher's pop time, and the runner's time to second base. To avoid bias in evaluating these roles, I adopted a data-driven approach to determine which factors make Elly De La Cruz successful in stealing second base. Using the collected data, I then sought ways to contain his stolen base attempts.

The first step in this process was to determine which statistics best capture the factors involved in a stolen base attempt. To identify these measures, I combined Baseball Savant baserunning, catching, and pitching season-level data. I focused specifically on Elly De La Cruz's stolen base attempts from the 2023 and 2024 MLB seasons. The key variable I used to estimate the probability of a successful steal was the runner's jump. While some define "jump" as the speed of a runner's reaction to a pitcher's movement, I defined it as the separation distance between the pitcher's initial motion and the release of the pitch. Measuring this separation allowed me to incorporate pitcher time to the plate into the analysis and account for its influence on stolen base outcomes.

## 2. Results and Analysis

### 2.1 Pitch Velocity

The first graph examined the relationship between pitch velocity, lead distance from first base, and stolen base outcome. No strong correlation was observed between velocity and lead distance. However, the data suggests that De La Cruz is extremely likely to successfully steal second base when pitch velocity is below 85 mph.

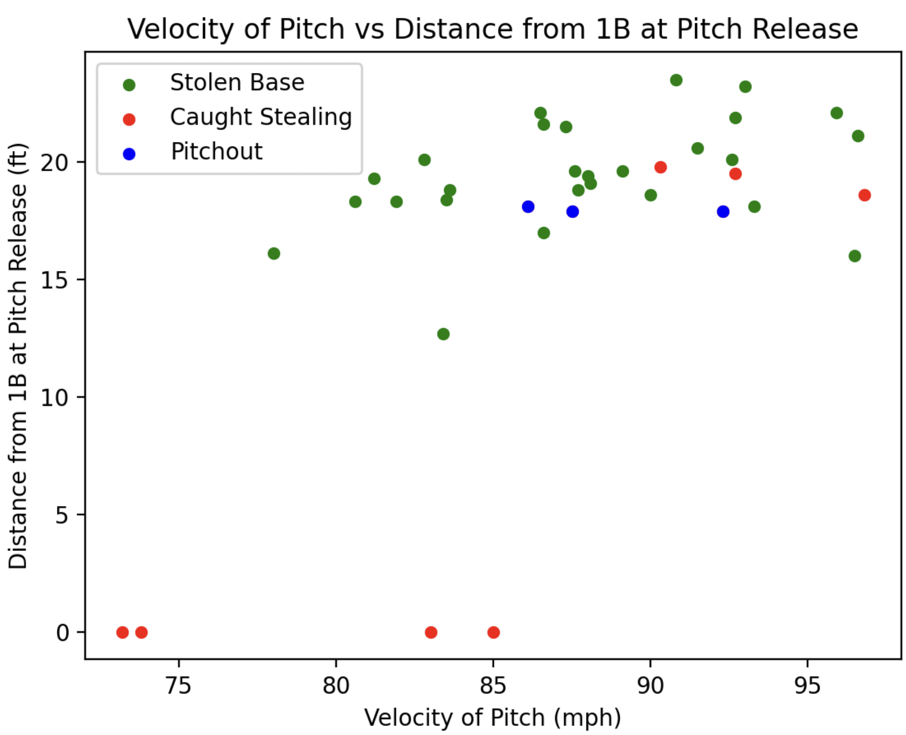


Figure 1: Velocity of Pitch vs. Distance from First Base at Pitch Release

From Figure 1, it is visible that there is no clear relationship between pitch velocity and separation from first base with the stolen base outcome. The green dots indicate successful stolen base attempts, the red dots represent times when De La Cruz was caught stealing, and the blue dots mark intentional pitchouts. Importantly, the blue points do not mean he was automatically out, but rather that a pitchout was attempted, and in some cases he still succeeded. Data points where his distance from first base at pitch release is zero indicate pickoffs by the pitcher. With this data, we can suggest that it is extremely likely for De La Cruz to successfully steal second base when the pitch velocity is under 85 mph.

### 2.2 Pitcher Handedness

To evaluate whether pitcher handedness influences stolen base outcomes, I examined the relationship between lead distance at pitch release and the success of the steal attempt. I initially hypothesized that De La Cruz would be more likely to be thrown out when facing left-handed pitchers, particularly if he delayed his move rather than breaking on the first motion. However, the data did not strongly support this claim. Instead, the results indicate that De La Cruz exhibited more variability in his leads against left-handed pitchers compared to right-handed pitchers. This suggests that while handedness may affect his approach to taking a lead, it does not directly determine whether he is successful in stealing second base.

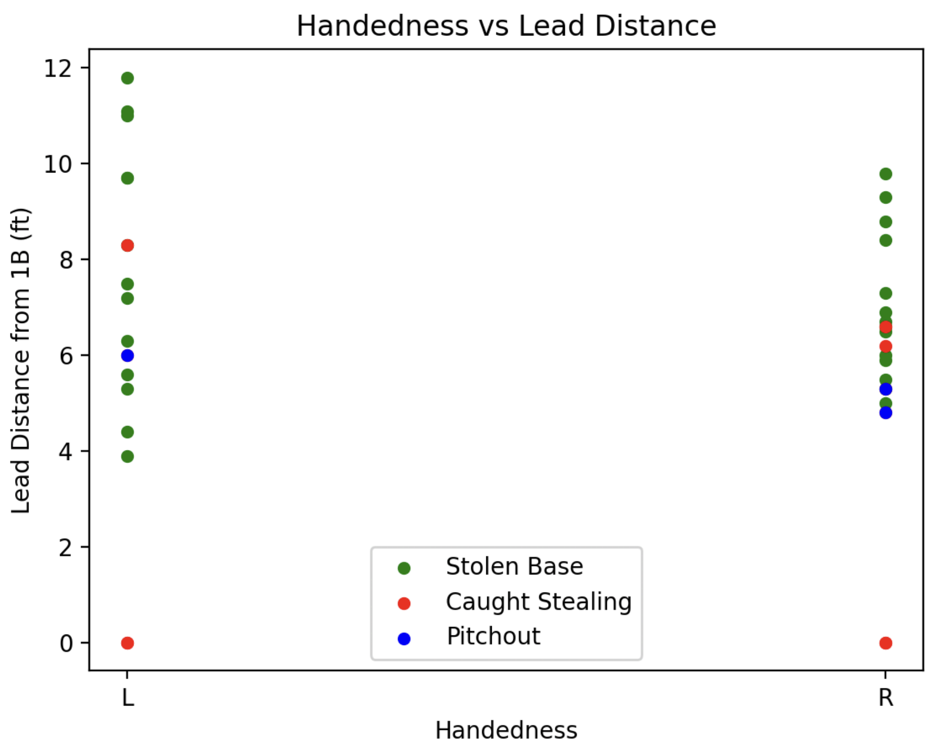


Figure 2: Handedness vs Lead Distance

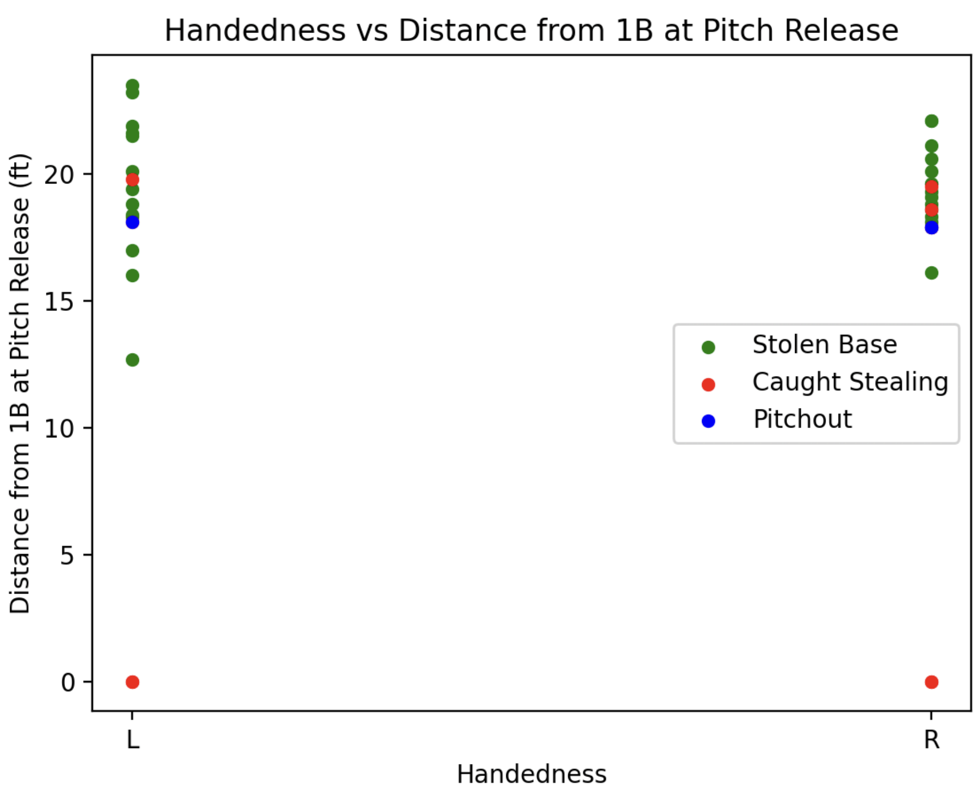

Figure 3: Handedness vs Distance from 1B at Pitch Release

**2.3 Runner's Jump**

Lastly, I explored whether the jump that Elly De La Cruz takes off first base relates to his stolen base outcome. Much like the previous analyses, the results did not provide a clear indication that jump distance was directly correlated with success or failure. While jump is often considered a critical factor in stolen base attempts, the data here suggests that De La Cruz's elite speed and instincts may overshadow the influence of jump alone.

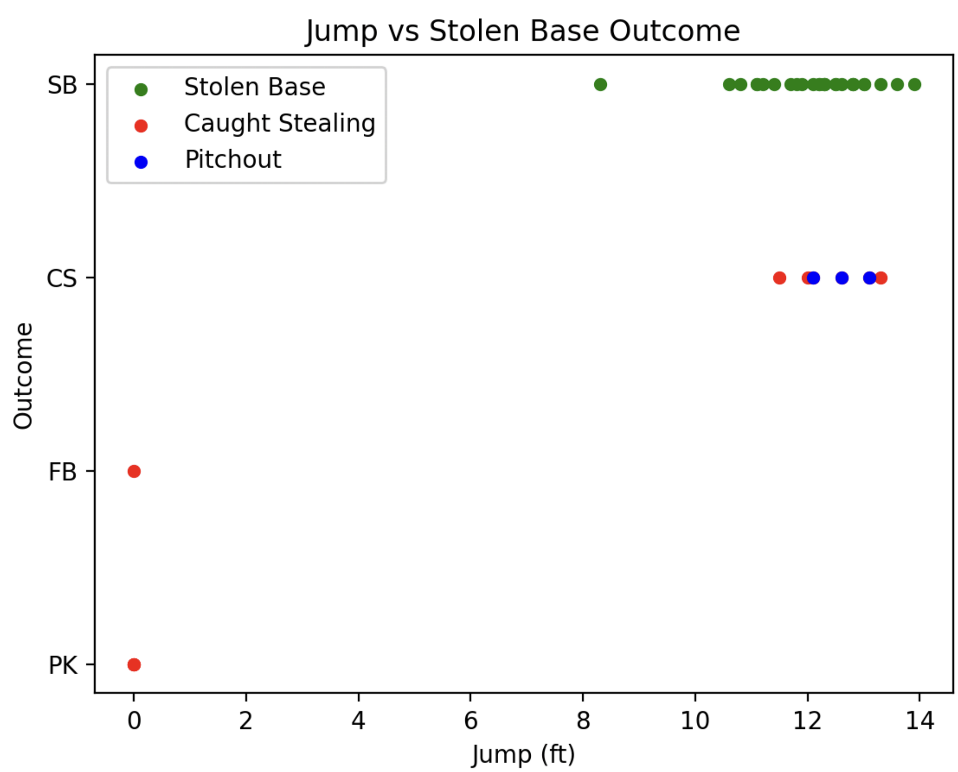


Figure 4: Jump vs Stolen Base Outcome

## 3. Discussion and Conclusion

This project examined several variables, pitch velocity, pitcher handedness, catcher pop time, and runner jump, to determine whether Elly De La Cruz's stolen base success could be predicted or contained. Across these analyses, no single factor consistently explained his outcomes, instead the findings highlight the complexity of stolen base attempts, where multiple small advantages combine to favor the runner.

The next step is to expand this analysis beyond De La Cruz by incorporating league wide play by play data. Once the 2023 dataset is available, I plan to retrieve all stolen base attempts across Major League Baseball and compare them with De La Cruz's outcomes. This comparison will provide a broader context for evaluating whether his success is unique or simply an extreme case of league wide trends. I will then apply the same code logic to the 2024 dataset, while noting that differences between the two seasons, particularly the adjustment period following the introduction of the pitch clock and pickoff restrictions, may influence the results.

Ultimately, the goal is to determine whether it is truly possible to stop one of the fastest players in the league from taking an extra 90 feet. By situating De La Cruz's performance within the larger environment of evolving MLB rules and strategies, this project aims to provide a more comprehensive understanding of stolen base dynamics and the challenges of defending against elite baserunners.

### 3.1 Future Works

Future work will involve applying statistical tests, such as p values, to determine whether we can reject the null hypothesis regarding the influence of these variables on stolen base outcomes. Additionally, rather than relying on visual inspection, I plan to implement machine learning methods, such as logistic regression, to model stolen base success more formally. While this project served as a valuable introduction to baseball data analysis, these next steps will provide stronger evidence and more generalizable insights.

## 4. Works Cited

MLB Advanced Media. *Statcast Data.* Baseball Savant, Major League Baseball, https://baseballsavant.mlb.com. Accessed 4 Oct. 2024.